



Independent QCE practice

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Practice Set A

2025 SYLLABUS EDITION

Question and response book

Chemistry

Paper 2

Time allowed

- Perusal time - 5 minutes
- Working time - 90 minutes

General instructions

- Answer all questions in this question and response book.
- A QCAA-approved calculator is permitted.
- The Chemistry formula and data book is provided.
- Planning paper will not be marked.

Written response (55 marks)

- 9 questions
- Show all reasoning and working.

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Written response

Instructions

- Write using black or blue pen.
 - Questions worth more than one mark require chemical reasoning and/or working to support the answer.
 - Give units, state symbols, significant figures and chemical notation where required.
 - If you need more space, use an additional page and identify the question number.
 - This paper is worth 55 marks.
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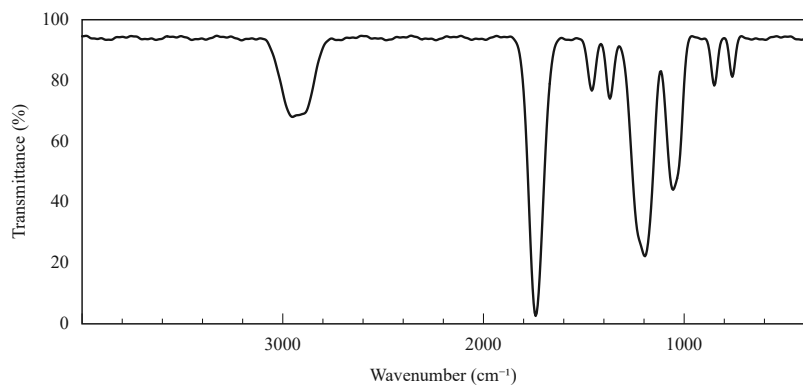
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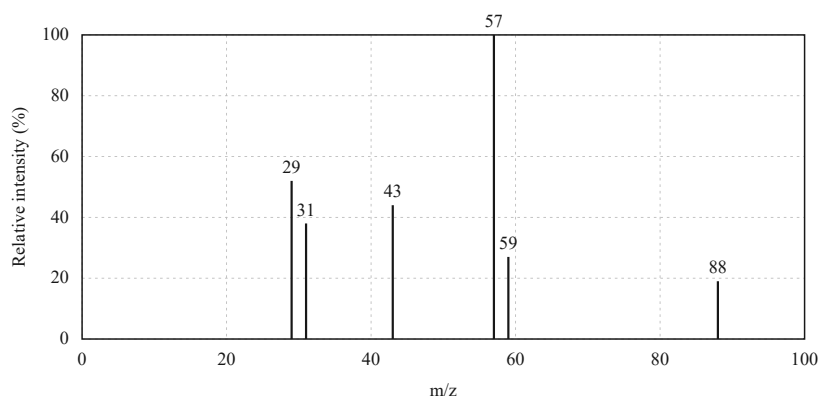
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QUESTION 1 (6 marks)

An organic liquid has empirical formula C_2H_4O . It was produced by reacting methanol with a carboxylic acid. Its infrared and mass spectra are shown.



Infrared spectrum



Mass spectrum

- a) Identify the compound class and justify your answer using the infrared spectrum.

[2 marks]

QUESTION 1 - CONTINUED

b) Use the molecular-ion peak to determine the molecular formula.

[2 marks]

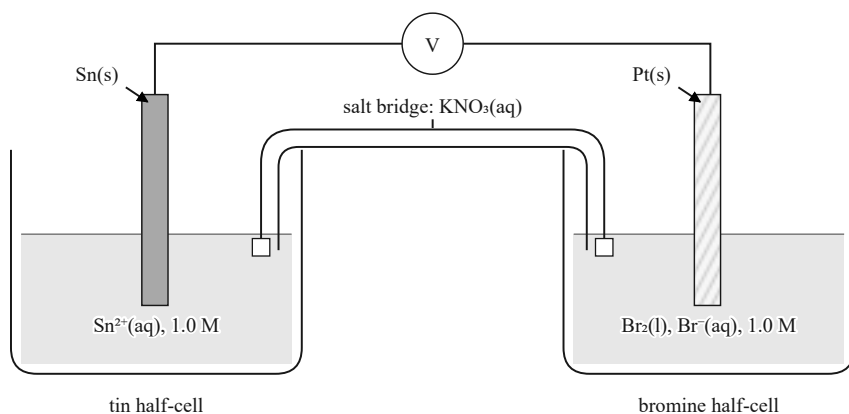
c) Deduce the structural formula and IUPAC name of the liquid. Explain how the synthesis information supports your answer.

[2 marks]

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QUESTION 2 (7 marks)

A galvanic cell uses a tin half-cell and a bromine/bromide half-cell under standard conditions. The inert electrode in the bromine half-cell is platinum. The standard reduction potentials are $E^\circ(\text{Br}_2/\text{Br}^-) = +1.08 \text{ V}$ and $E^\circ(\text{Sn}^{2+}/\text{Sn}) = -0.14 \text{ V}$.



Tin–bromine galvanic cell

- a) Write the oxidation and reduction half-equations. Include states. [2 marks]

- b) State the direction of electron flow and the direction in which anions from the salt bridge initially move. [2 marks]

QUESTION 2 - CONTINUED

c) Calculate the standard cell potential.

[1 mark]

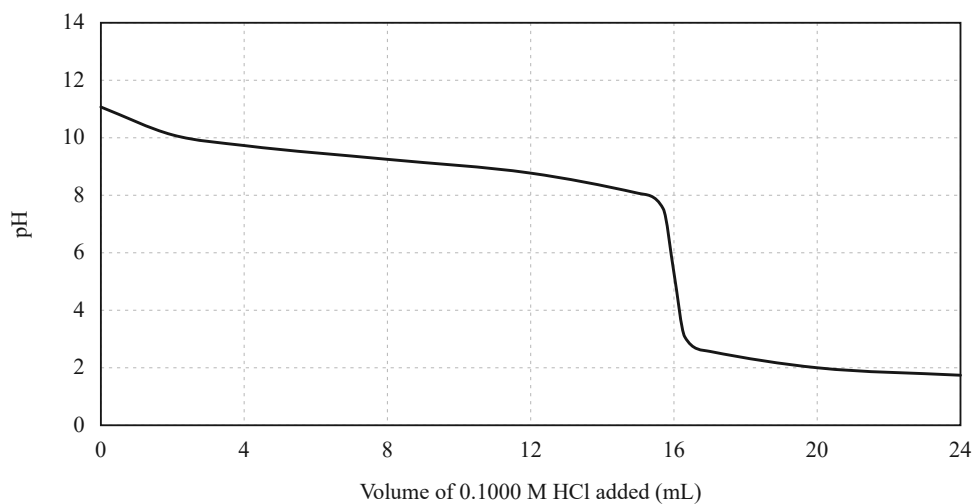
d) Describe two observable changes as the cell operates.

[2 marks]

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QUESTION 3 (6 marks)

At 25 °C, 20.00 mL of an ammonia solution is titrated with 0.1000 M hydrochloric acid. The titration curve has an equivalence volume of 16.00 mL and a pH of 9.25 at half-equivalence.



Ammonia–hydrochloric acid titration curve

Indicator	Transition range
methyl red	4.4-6.2
bromothymol blue	6.0-7.6
phenolphthalein	8.3-10.0

Indicator transition ranges

- a) Calculate the initial concentration of ammonia.

[2 marks]

- b) Use the half-equivalence point to determine K_b for ammonia.

[2 marks]

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QUESTION 3 - CONTINUED

c) Select the most suitable indicator from the table.

[1 mark]

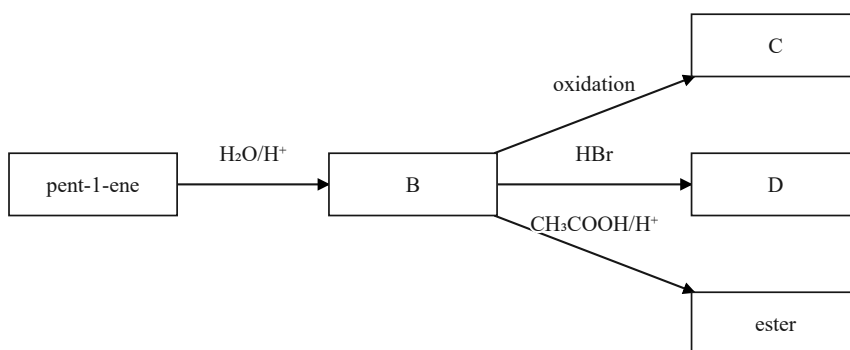
d) The burette was rinsed with water immediately before it was filled with hydrochloric acid. State whether using the labelled acid concentration makes the calculated ammonia concentration too high or too low.

[1 mark]

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QUESTION 4 (7 marks)

Use the displayed reaction pathway, which starts with pent-1-ene. Compound B is the major hydration product.



Reaction pathway

- a) Draw the structural formula of B and give its IUPAC name.

[2 marks]

- b) State a reagent for the conversion of B to C and draw the structural formula of C.

[2 marks]

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QUESTION 4 - CONTINUED

- c) Classify the conversion of pent-1-ene to B and the conversion of B to D.

[2 marks]

- d) Draw the ester formed when B reacts with ethanoic acid.

[1 mark]

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QUESTION 5 (6 marks)

0.800 mol of $\text{PCl}_5(\text{g})$ is placed in a 2.00 L vessel. At equilibrium, $[\text{Cl}_2] = 0.150 \text{ M}$ for $\text{PCl}_5(\text{g}) \rightleftharpoons \text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g})$.

- a) Determine the equilibrium concentrations of PCl_5 and PCl_3 . [2 marks]

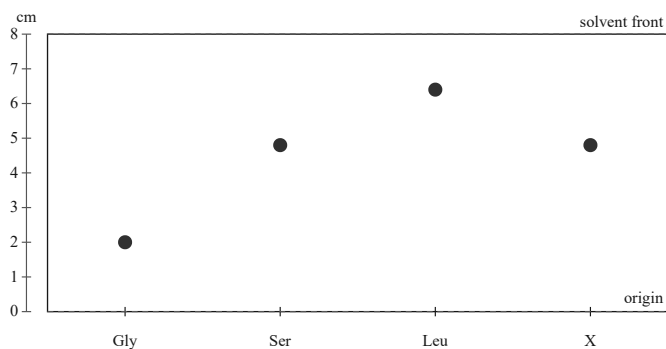
- b) Calculate K_c . [2 marks]

- c) At the same temperature, a second mixture has $[\text{PCl}_5] = 0.200 \text{ M}$, $[\text{PCl}_3] = 0.100 \text{ M}$ and $[\text{Cl}_2] = 0.300 \text{ M}$. Infer the direction of the net reaction. [2 marks]

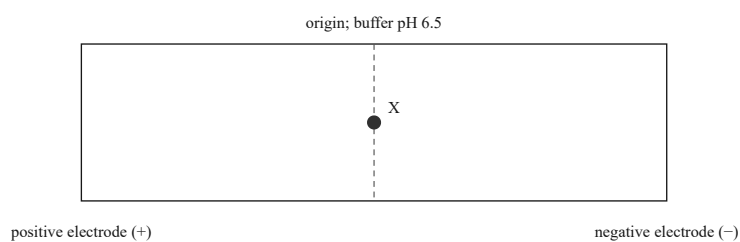
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QUESTION 6 (6 marks)

An unknown amino acid X is analysed using the paper chromatogram and electrophoresis arrangement shown. The chromatography uses polar cellulose paper with adsorbed water as the stationary phase and a less-polar organic solvent as the mobile phase. Serine has an isoelectric point of 5.68.



Paper chromatogram



Electrophoresis at pH 6.50

- a) Calculate the R_f value of X and identify it.

[2 marks]

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QUESTION 6 - CONTINUED

- b) Leucine moves further than glycine in the stated paper-chromatography system.
Explain this difference in terms of the stationary and mobile phases.

[2 marks]

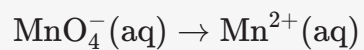
- c) Predict the net charge of X and its direction of movement in the pH 6.50 buffer.

[2 marks]

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QUESTION 7 (6 marks)

Acidified permanganate ions oxidise iron(II) ions to iron(III) ions. A 25.00 mL iron(II) sample requires 18.60 mL of 0.02000 M permanganate solution for complete reaction.



Reduction skeleton

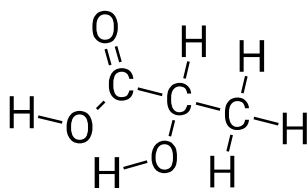
- a) Write the oxidation half-equation and the balanced reduction half-equation in acidic conditions. [3 marks]

- b) Calculate the concentration of iron(II) ions in the sample. [3 marks]

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QUESTION 8 (6 marks)

The displayed lactic acid monomer can form polylactic acid (PLA) by condensation polymerisation.



Lactic acid monomer

- a) Draw the repeating unit of PLA and identify the linking functional group. [2 marks]

- b) Write an equation for two lactic acid molecules forming a dimer. Show the small molecule eliminated. [2 marks]

- c) Explain why PLA is more susceptible to hydrolysis and biodegradation than polyethene. [2 marks]

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QUESTION 9 (5 marks)

Four Haber-process trials begin with the same reactant ratio. The percentage ammonia at equilibrium and the time taken to approach equilibrium are recorded.

Trial	Temperature (°C)	Pressure (atm)	Catalyst	NH ₃ at equilibrium	Time to approach equilibrium (min)
A	400	100	yes	25%	20
B	450	100	yes	18%	12
C	450	200	yes	28%	8
D	450	200	no	28%	65

Haber-process trial data

- a) Use trials A and B to infer whether the forward reaction is exothermic or endothermic. Explain your reasoning. [2 marks]

- b) Use trials C and D to explain the effect of the catalyst on rate and equilibrium yield. [2 marks]

- c) Identify the condition change responsible for the higher equilibrium ammonia percentage in trial C than trial B. [1 mark]

END OF PAPER

ADDITIONAL PAGE FOR STUDENT RESPONSES

Write the question number you are responding to.

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